

Trace Format Example

event	time	from node	to node	pkt type	pkt size	flags	fid	src addr	dst addr	seq num	pkt id
-------	------	--------------	------------	-------------	-------------	-------	-----	-------------	-------------	------------	-----------

```

r : receive (at to_node)
+ : enqueue (at queue)          src_addr : node.port (3.0)
- : dequeue (at queue)          dst_addr : node.port (0.0)
d : drop      (at queue)

```

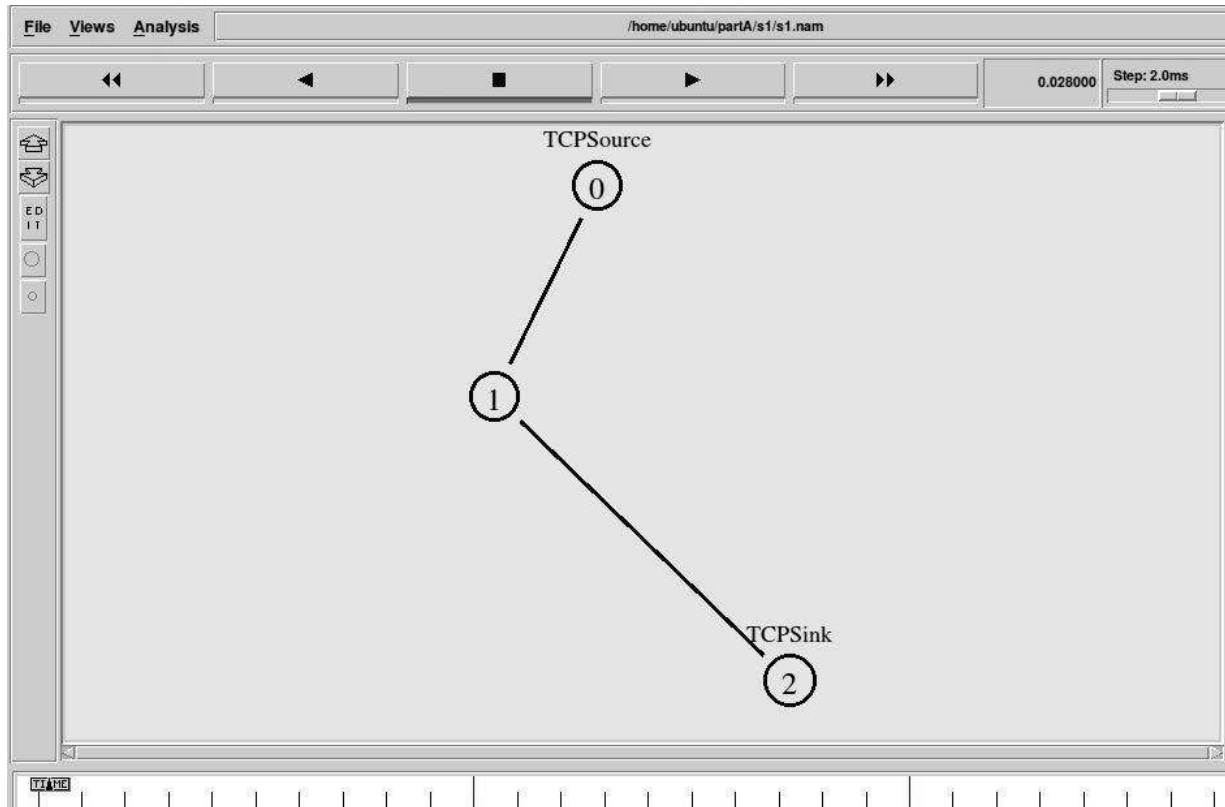
```

r 1.3556 3 2 ack 40 ----- 1 3.0 0.0 15 201
+ 1.3556 2 0 ack 40 ----- 1 3.0 0.0 15 201
- 1.3556 2 0 ack 40 ----- 1 3.0 0.0 15 201
r 1.35576 0 2 tcp 1000 ----- 1 0.0 3.0 29 199
+ 1.35576 2 3 tcp 1000 ----- 1 0.0 3.0 29 199
d 1.35576 2 3 tcp 1000 ----- 1 0.0 3.0 29 199
+ 1.356 1 2 cbr 1000 ----- 2 1.0 3.1 157 207
- 1.356 1 2 cbr 1000 ----- 2 1.0 3.1 157 207

```

1. An event (+, -, d, r) descriptor
2. Simulation time (in seconds) of that event
3. From node
4. To node (3,4 identifies the link on which the event occurred)
5. Packet type (in Bytes)
6. Packet size (in Bytes)
7. Flags (appeared as "-----" since no flag is set)
8. Flow id (fid) of IPv6 that a user can set for each flow at the input OTcl script. Even though fid field may not be used in a simulation, users can use this field for analysis purposes. The fid field is also used when specifying stream color for the NAM display.
9. Source address in forms of "**node.port**".
10. Destination address in forms of "**node.port**".
11. Network layer protocol's packet sequence number. Note that even though UDP implementations do not use sequence number, NS keeps track of UDP packet sequence number for analysis purposes.
12. Unique id of the packet

1. Simulate a three nodes point – to – point network with duplex links between them. Set the queue size and vary the bandwidth and find the number of packets dropped.



s1.awk

```
BEGIN{
    count = 0;
}
{
    event = $1;
    if(event == "d"){count++;}
}
END{
    printf("\nNumber of packets dropped is: %d\n", count);
}
```

s1.tcl***#create new simulation instance***

```
set ns [new Simulator]
```

#open trace file

```
settracefile [open s1.tr w]
$ns trace-all $tracefile
```

#open nam:animation file

```
setnamfile [open s1.nam w]
$ns namtrace-all $namfile
```

#define finish procedure to perform at the end of simulation

```
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
```

#dump all traces and close files

```
close $tracefile
close $namfile
```

#execute nam animation file

```
execnam s1.nam &
```

#execute awk file in background

```
execawk -f s1.awk s1.tr &
exit 0
```

```
}
```

#create 3 nodes

```
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
```

#create labels

```
$n0 label "TCPSource"
$n2 label "TCPSink"
```

#set color

```
$ns color 1 red
```

#create link between nodes /create topology

```
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
$ns duplex-link $n1 $n2 1Mb 10ms DropTail
```

#set queue size of N packets between n1 and n2

```
$ns queue-limit $n1 $n2 5
```

#create TCP agent and attach to node 0

```
settcp [new Agent/TCP]
$ns attach-agent $n0 $tcp
```

#create TCP sink agent and attach to node 2

```
settcpsink [new Agent/TCPSink]
$ns attach-agent $n2 $stepsink
```

#create traffic: FTP: create FTP source agent on top of TCP and attach to TCP agent

```
set ftp [new Application/FTP]
$ftp attach-agent $tcp
```

#connect TCP agent with TCP sink agent

```
$ns connect $tcp $stepsink
```

#set the color

```
$tcp set class_ 1
```

#schedule events

```
$ns at 0.2 "$ftp start"
$ns at 2.5 "$ftp stop"
$ns at 3.0 "finish"
$ns run
```

Output

```
#gedit s1.tcl
#sudo ns s1.tcl
```

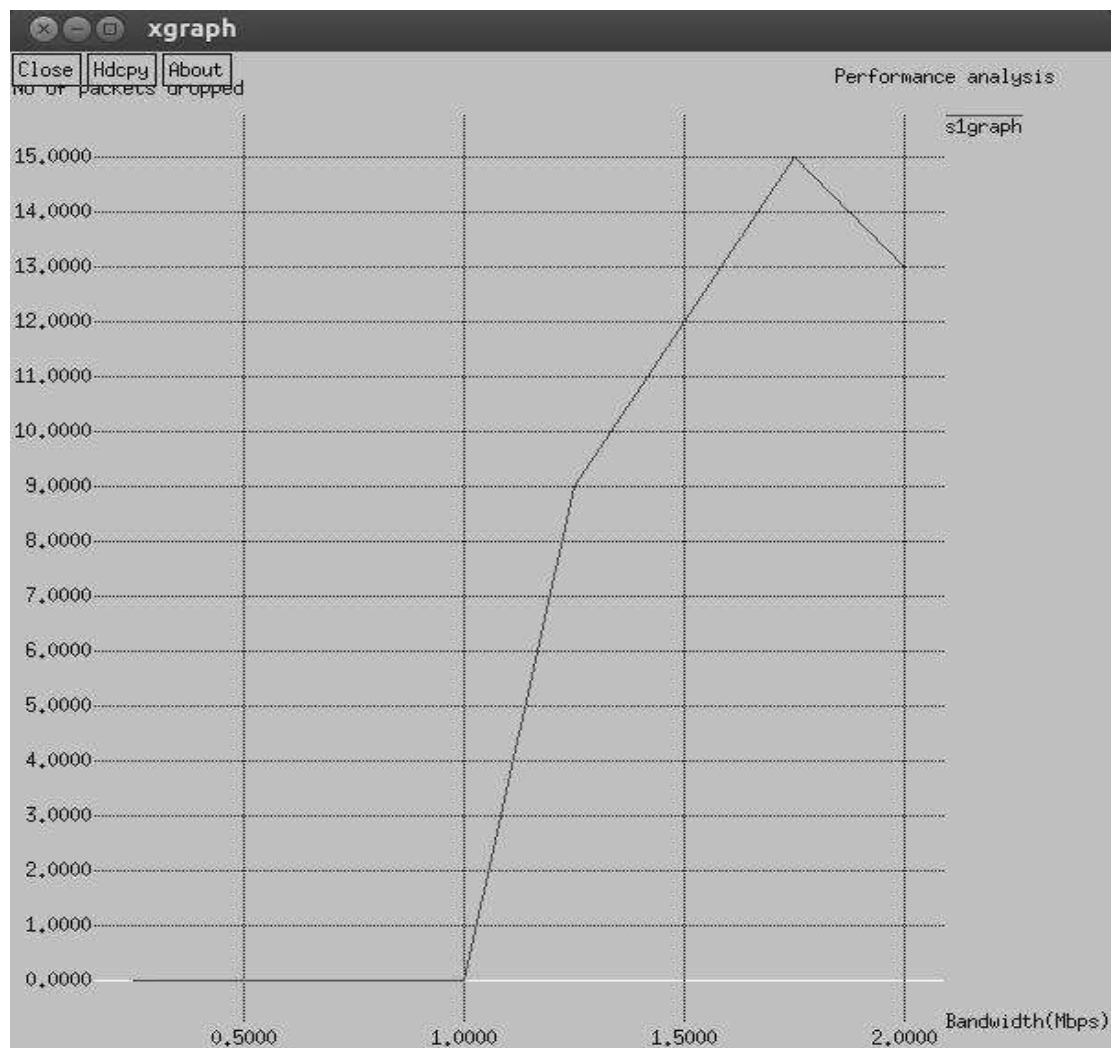
Number of packets dropped is: 0

```
#gedit s1graph
```

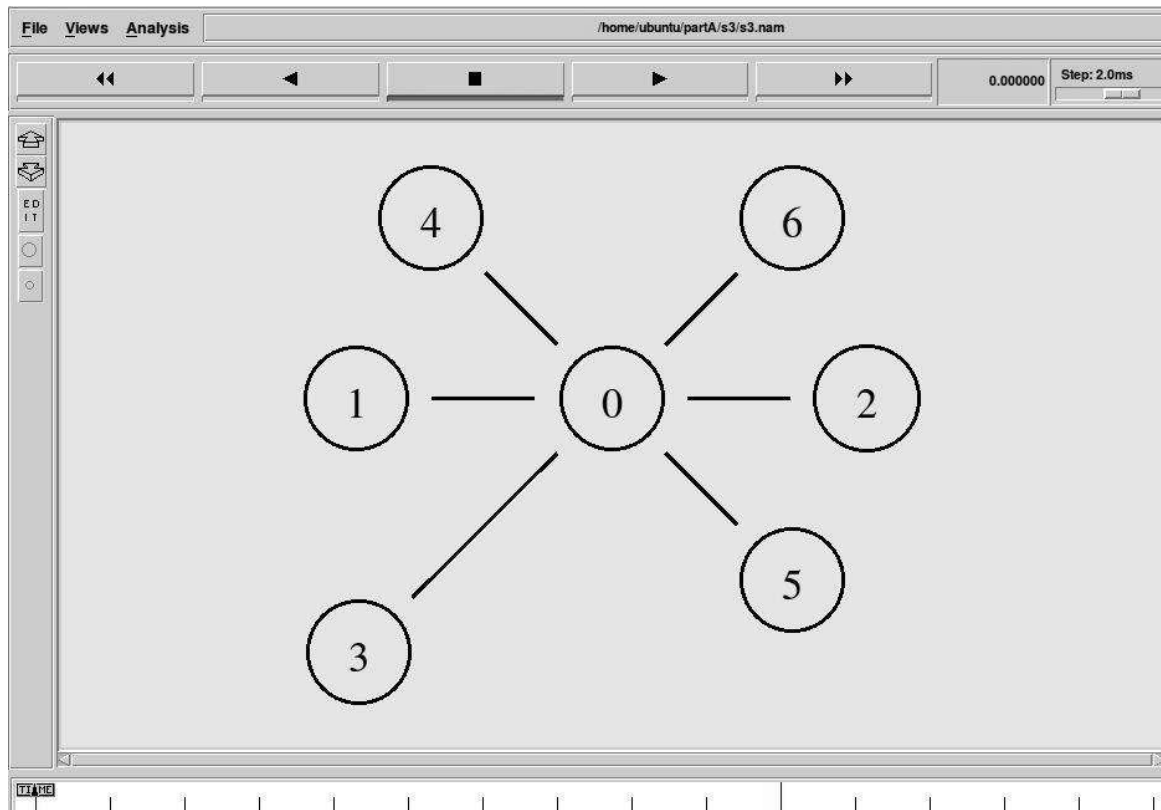
s1graph

0.25	0
0.50	0
0.75	0
1.00	0
1.25	9
1.50	12
1.75	15
2.00	13

```
#xgraph -x "Bandwidth(Mbps)" -y "No of packets dropped" -t "Performance analysis" s1graph
```



2. Simulate the transmission of ping messages over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion.



s2.awk

```
BEGIN {  
    count = 0;  
}  
{  
    event = $1;  
    if(event == "d"){    count++;}  
}  
END {  
    printf("No of packets dropped : %d\n",count);  
}
```

s2.tcl

```
set ns [new Simulator]
```

```
setnamfile [open s2.nam w]
```

```
$ns namtrace-all $namfile
```

```
settracefile[open s2.tr w]
```

```
$ns trace-all $tracefile
```

```
proc finish {} {  
    global ns namfiletracefile  
    $ns flush-trace  
    close $namfile  
    close $tracefile  
    execnam s2.nam &  
    execawk -f s2.awk s2.tr &  
    exit 0  
}
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
set n6 [$ns node]
```

```
$ns duplex-link $n1 $n0 1Mb 10ms DropTail
```

```
$ns duplex-link $n2 $n0 1Mb 10ms DropTail
```

```
$ns duplex-link $n3 $n0 1.75Mb 20ms DropTail
```

```
$ns duplex-link $n4 $n0 1Mb 10ms DropTail
```

```
$ns duplex-link $n5 $n0 1Mb 10ms DropTail
```

```
$ns duplex-link $n6 $n0 1Mb 10ms DropTail
```

```
$ns duplex-link-op $n0 $n1 orient right
```

```
$ns duplex-link-op $n0 $n2 orient left
```

```
$ns duplex-link-op $n0 $n3 orient right-up
```

```
$ns duplex-link-op $n0 $n4 orient right-down
```

```
$ns duplex-link-op $n0 $n5 orient left-up
```

```
$ns duplex-link-op $n0 $n6 orient left-down
```

```
Agent/Ping instprocrecv {from rtt} {
```

```
    $self instvar node_
```

```
    puts "node [$node_ id] received ping answer from $from with round-trip-time $rttms"
```

```
}
```

```
set p1 [new Agent/Ping]
set p2 [new Agent/Ping]
set p3 [new Agent/Ping]

set p4 [new Agent/Ping]
set p5 [new Agent/Ping]
set p6 [new Agent/Ping]

$ns attach-agent $n1 $p1
$ns attach-agent $n2 $p2
$ns attach-agent $n3 $p3
$ns attach-agent $n4 $p4
$ns attach-agent $n5 $p5
$ns attach-agent $n6 $p6

$ns queue-limit $n0 $n4 1
$ns queue-limit $n0 $n5 2
$ns queue-limit $n0 $n6 2

$ns connect $p1 $p4
$ns connect $p2 $p5
$ns connect $p3 $p6
$ns at 0.2 "$p1 send"
$ns at 0.4 "$p2 send"
$ns at 0.6 "$p3 send"
$ns at 1.0 "$p4 send"
$ns at 1.2 "$p5 send"
$ns at 1.4 "$p6 send"
$ns at 2.0 "finish"
$ns run
```

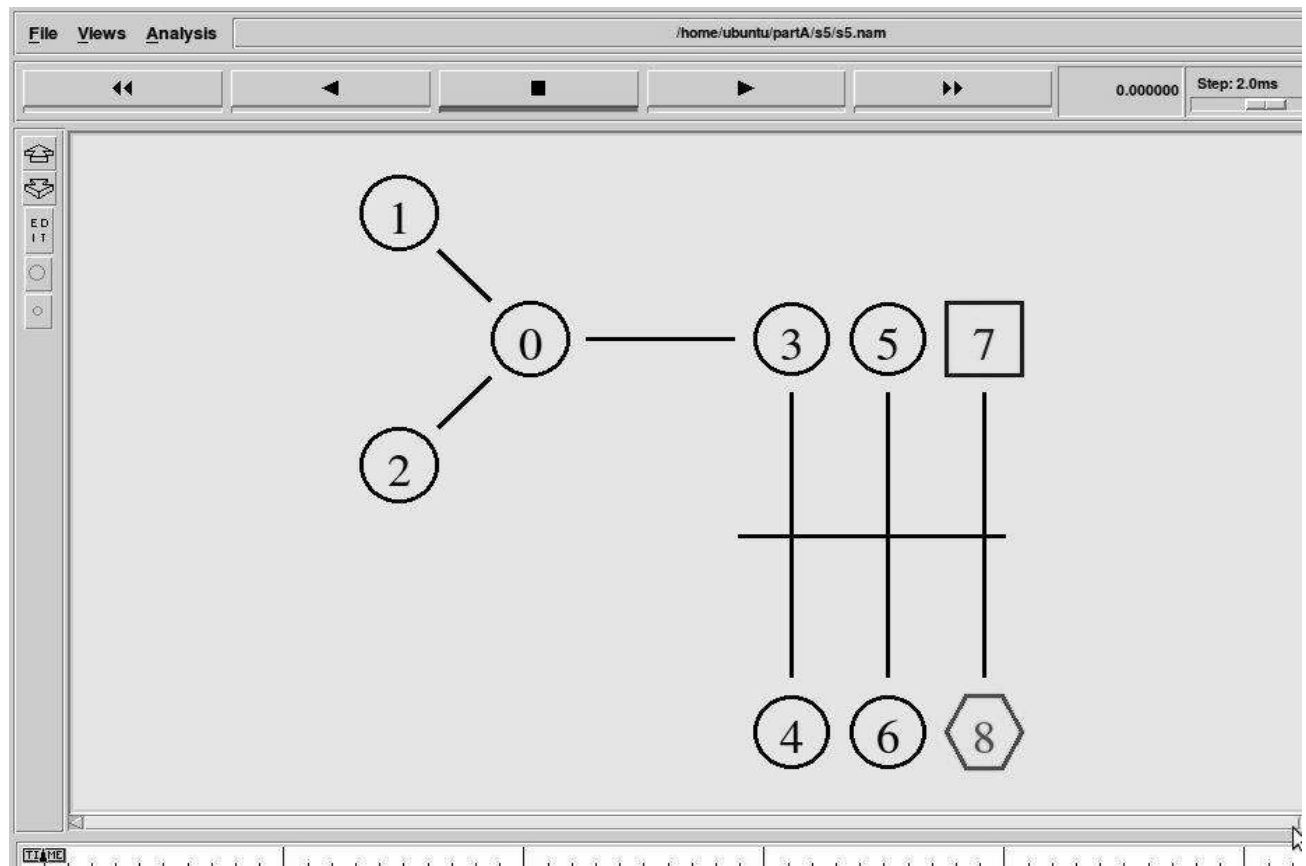
Output

```
#gedit s2.tcl
#sudo ns s2.tcl
```

node 2 received ping answer from 5 with round-trip-time 42.0 ms.
node 3 received ping answer from 6 with round-trip-time 61.6 ms.
node 5 received ping answer from 2 with round-trip-time 42.0 ms.
node 6 received ping answer from 3 with round-trip-time 61.6 ms.

No of packets dropped : 2

3. Simulate an Ethernet LAN using n nodes and set multiple traffic nodes and plot congestion window for different source / destination.



s3.awk

```
BEGIN {
}
{
    if($6=="cwnd_")
    {
        printf("%f\t%f\n",$1,$7);
    }
}
END {
}
```

s3.tcl

```
set ns [new Simulator]
```

```
setnamfile [open s3.nam w]
$ns namtrace-all $namfile
```

```
settracefile [open s3.tr w]
$ns trace-all $tracefile
```

```
proc finish {} {
    global ns namfiletracefile
    $ns flush-trace
    close $namfile
    close $tracefile

    execnam s3.nam &
    exit 0
}
```

```
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
set n7 [$ns node]
set n8 [$ns node]
```

```
$ns color 1 Blue
$ns color 2 Red
```

```
$n7 shape box
$n7 color Blue
$n8 shape hexagon
$n8 color Red
```

```
$ns duplex-link $n1 $n0 2Mb 10ms DropTail
$ns duplex-link $n2 $n0 2Mb 10ms DropTail
$ns duplex-link $n0 $n3 1Mb 20ms DropTail
```

```
$ns make-lan "$n3 $n4 $n5 $n6 $n7 $n8" 512Kb 40ms LL Queue/DropTail Mac/802_3
```

```
$ns duplex-link-op $n1 $n0 orient right-down
$ns duplex-link-op $n2 $n0 orient right-up
$ns duplex-link-op $n0 $n3 orient right
```

```
$ns queue-limit $n0 $n3 20
```

```
set tcp1 [new Agent/TCP/Vegas]
$ns attach-agent $n1 $tcp1
```

```
set sink1 [new Agent/TCPSink]
$ns attach-agent $n7 $sink1
```

```
set ftp1 [new Application/FTP]
$ftp1 attach-agent $tcp1
```

```
$ns connect $tcp1 $sink1
```

```
$tcp1 set class_ 1
$tcp1 set packetSize_ 55
```

```
set tfile1 [open cwnd1.tr w]
$tcp1 attach $tfile1
$tcp1 trace cwnd_
```

```
set tcp2 [new Agent/TCP/Reno]
$ns attach-agent $n2 $tcp2
```

```
set sink2 [new Agent/TCPSink]
$ns attach-agent $n8 $sink2
```

```
set ftp2 [new Application/FTP]
$ftp2 attach-agent $tcp2
```

```
$ns connect $tcp2 $sink2
```

```
$tcp2 set class_ 2
$tcp2 set packetSize_ 55
```

```
set tfile2 [open cwnd2.tr w]
$tcp2 attach $tfile2
$tcp2 trace cwnd_
```

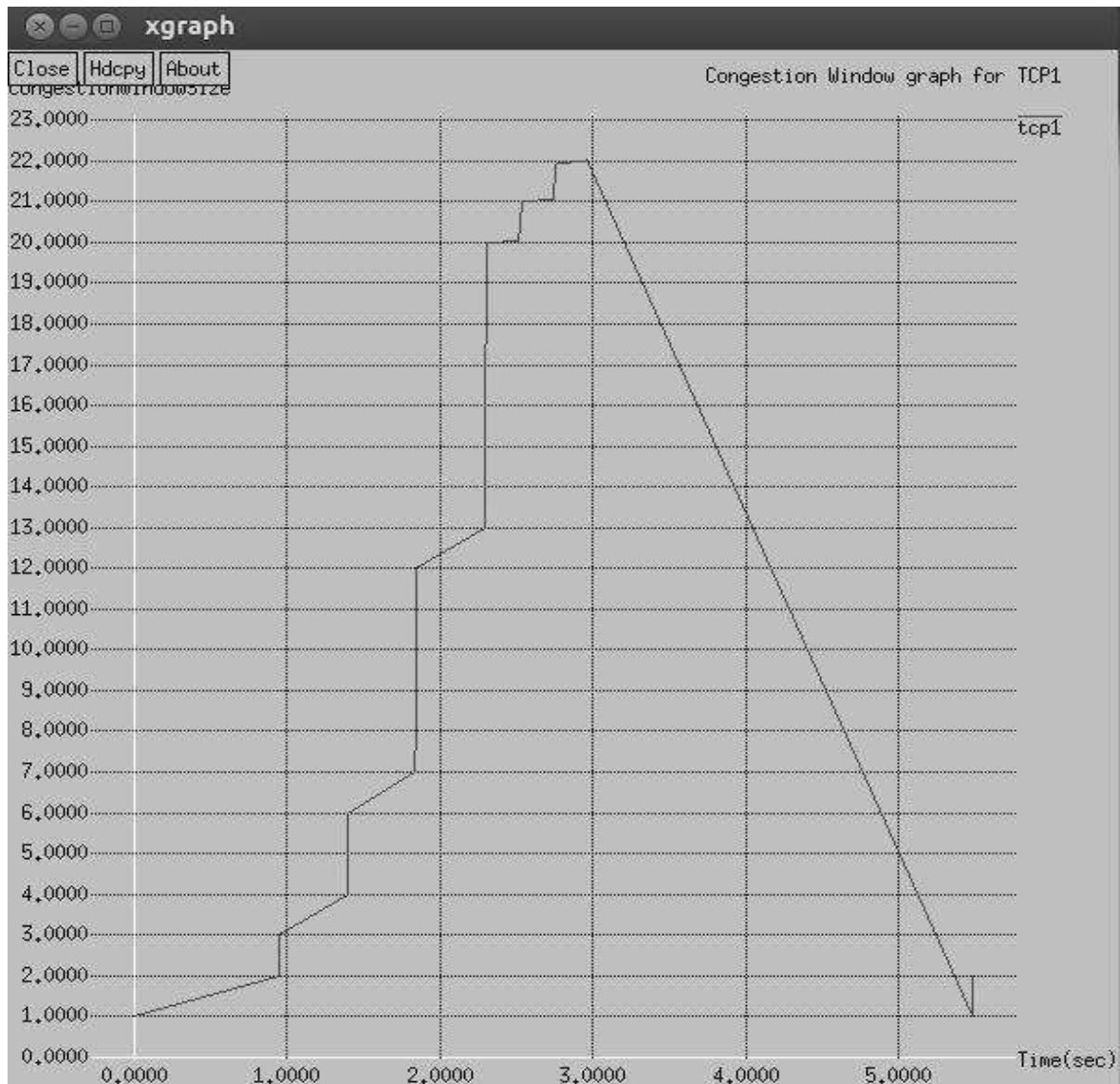
```
$ns at 0.5 "$ftp1 start"
$ns at 1.0 "$ftp2 start"
$ns at 5.0 "$ftp2 stop"
$ns at 5.0 "$ftp1 stop"
$ns at 5.5 "finish"
$ns run
```

Output

```
#gedit s3.tcl  
#sudo ns s3.tcl
```

```
#awk -f s3.awk cwnd1.tr >TCPVegas
```

```
#xgraph -x "Time(sec)" -y "CongestionWindowSize" -t "Congestion Window graph for TCP1" TCPVegas
```



```
#awk-f s3.awk cwnd2.tr>TCPReno
```

```
#xgraph -x "Time(sec)" -y "CongestionWindowSize" -t "Congestion Window graph for TCP2" TCPReno
```

